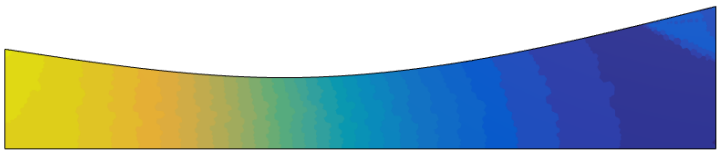
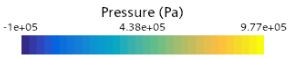
Simcenter STAR-CCM+







TP2 : MODELING A LAVAL NOZZLE ON STARCCM+

Fluids dynamics

ELS 2 - Mf421

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1-Analytical study

1. Calcul of the exit pressure

To calculate P_S we must use the following formula:

$$\frac{P_S}{P_0} = (1 + \frac{\gamma - 1}{2} M^2)^{\frac{-\gamma}{\gamma - 1}}$$

Then

$$P_S = P_0(1 + \frac{\gamma - 1}{2} M^2)^{\frac{-\gamma}{\gamma - 1}}$$

$$P_S = 1000000(1 + \frac{1.4 - 1}{2} 2.94^2)^{\frac{-1.4}{1.4 - 1}} = 29\ 795\ Pa$$

2. Calcul of the mass flow.

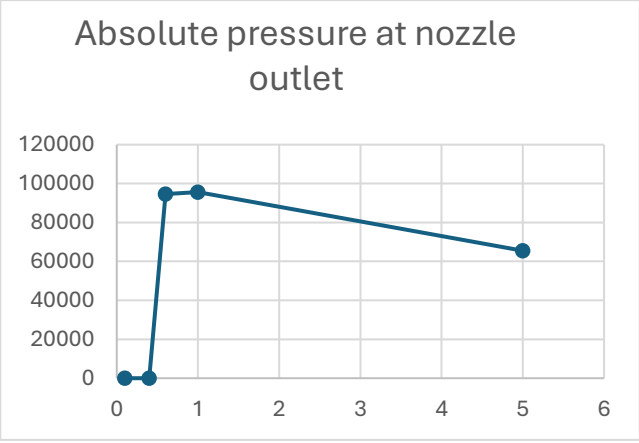
$$\dot{m} = P_0 A_c \sqrt{\frac{k}{RT_0} (\frac{2}{k + 1})^{\frac{k+1}{k-1}}}$$

The analytical result gives:

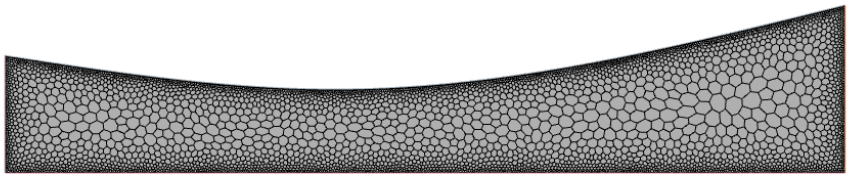
$$\dot{m} = 3.61\ kg/s$$

2-Mesh convergance study

Mesh	Base size (mm)	Absolute pressure at nozzle outlet
1	5	65453
2	1	95612
3	0,6	94578
4	0,4	Error
5	0,1	Error



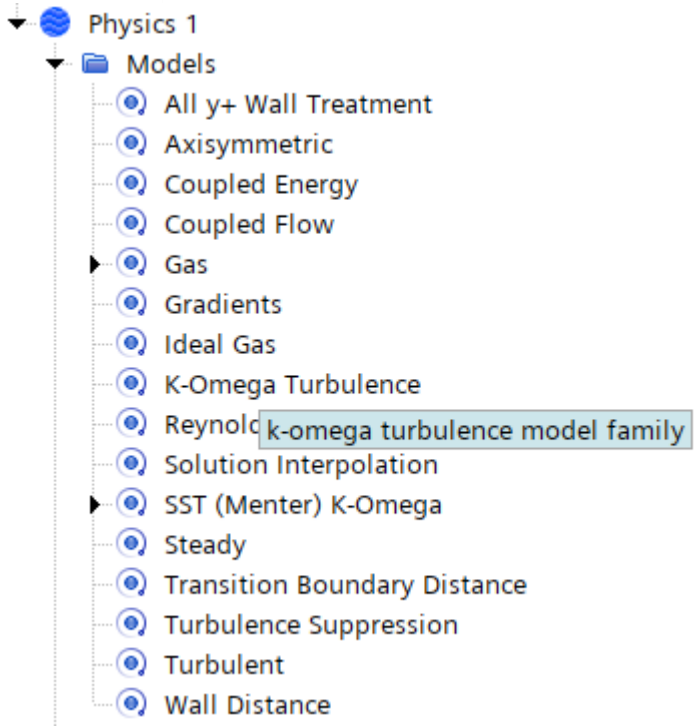
3-Actual model of mesh





Mesh base sier 0.6

4-Physics justification



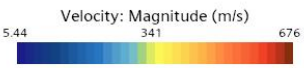
5- Study

1. Pressure outlet 300 kPa

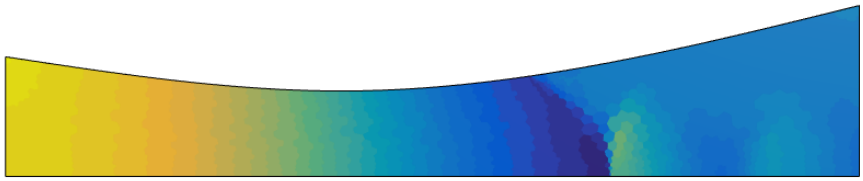
1) Flow Behavior

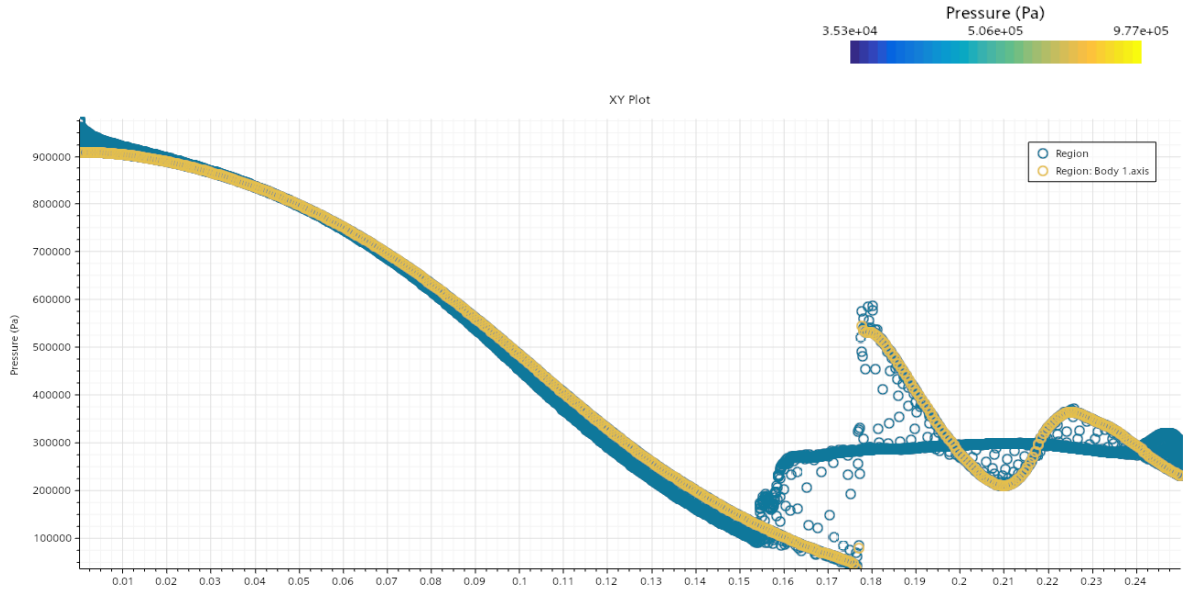




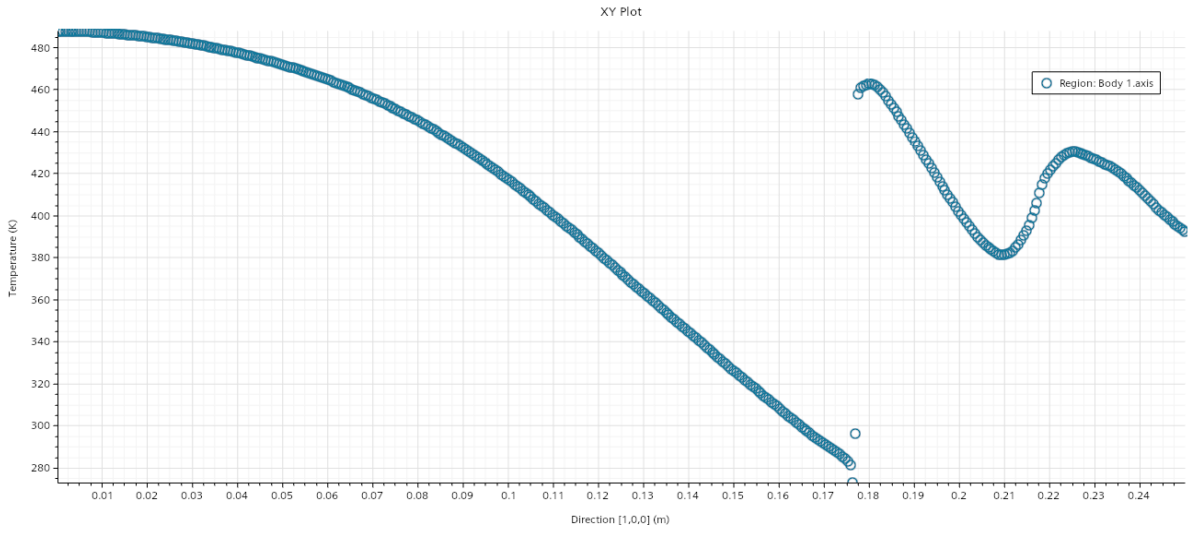


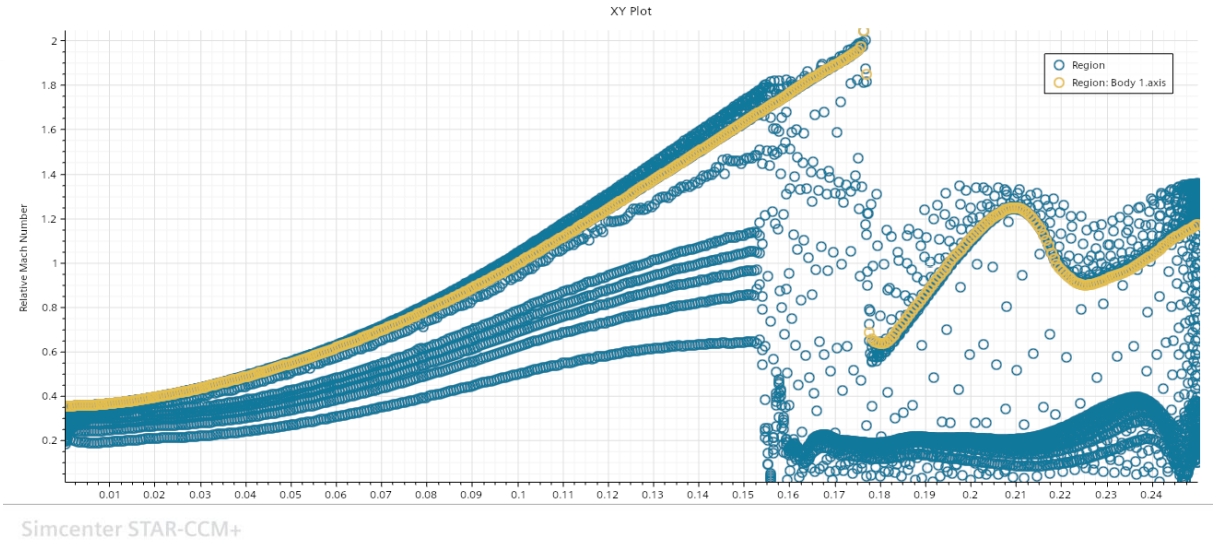
2) Pressure profile





3) Temperature and Mach number profile





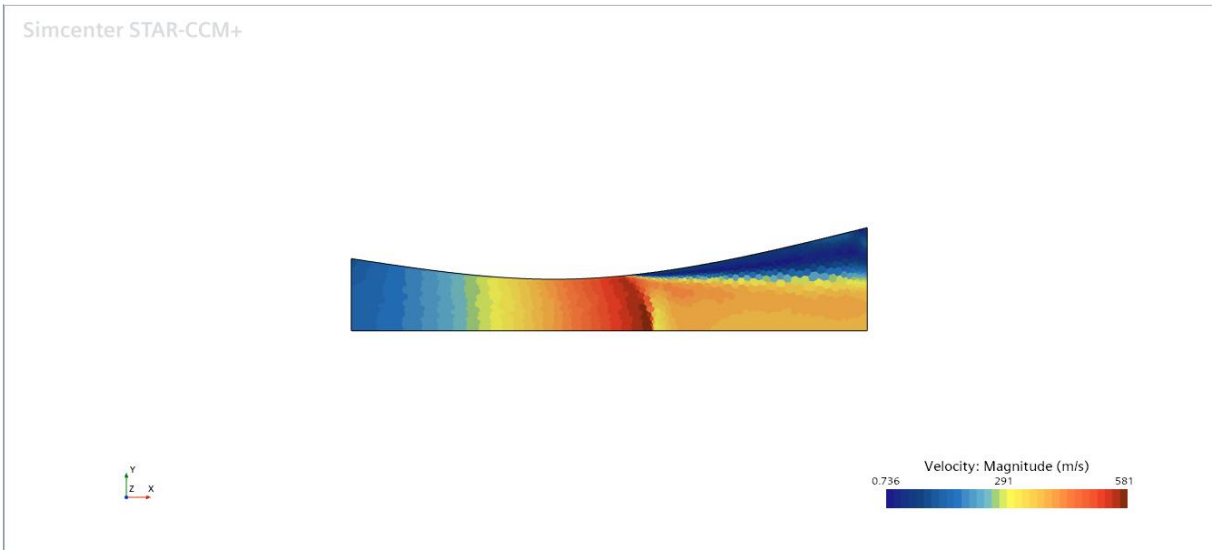




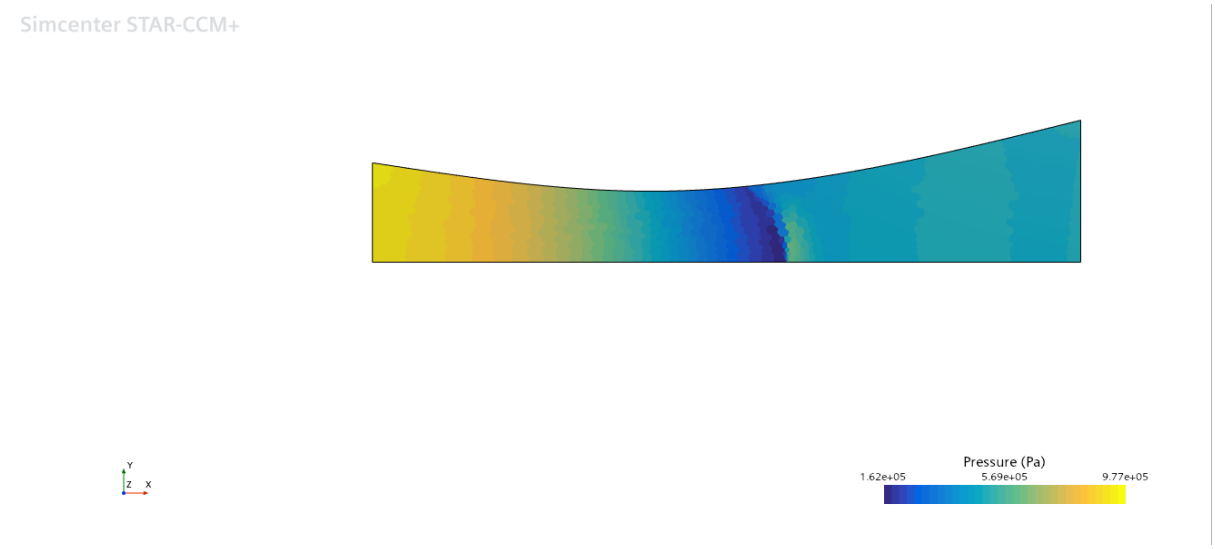
4) Comparison with analytical results

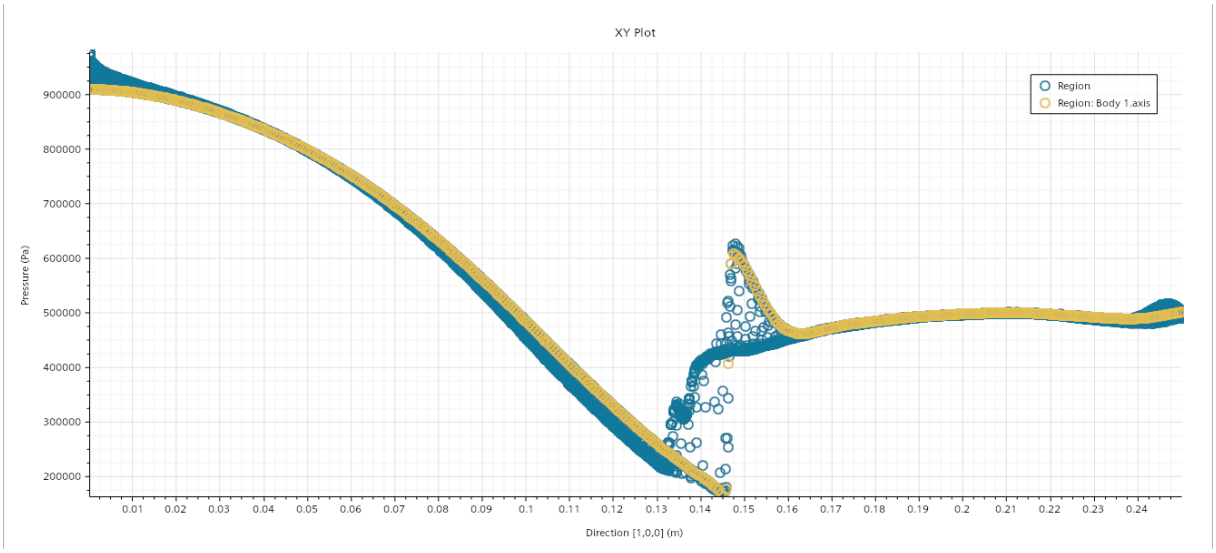
2. Pressure outlet 500kPa

1) Flow Behavior

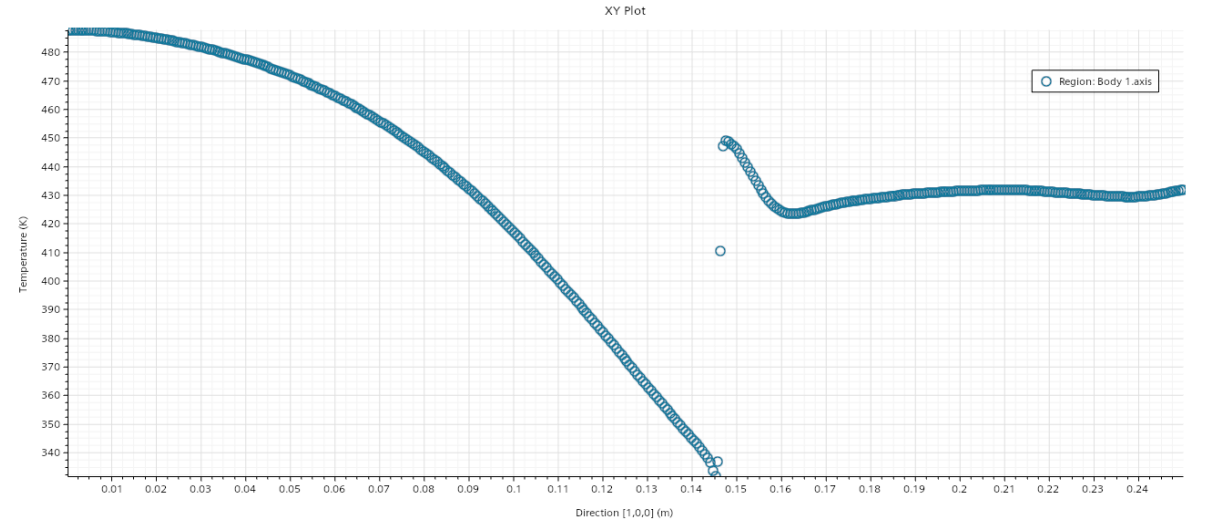


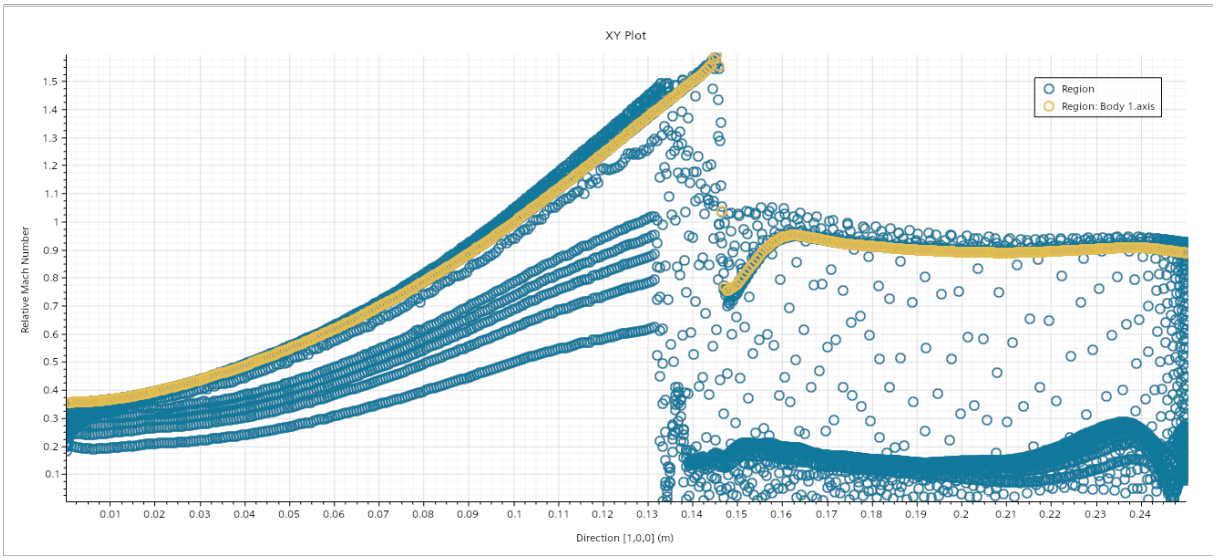
2) Pressure profile

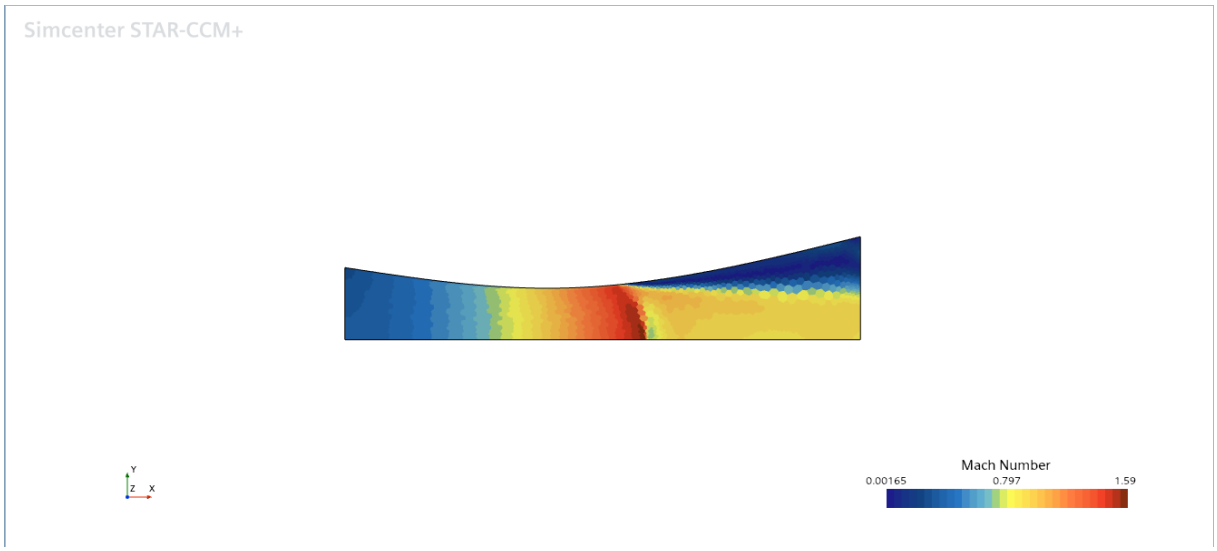




3) Temperature and Mach number profile







4) Comparison with analytical results